Harmony patterns: speech, suffixes and spelling

Turkish

Front/back
Productive suffix alternation

Rounding
High suffix vowels alternate
el-im, at-ım, yol-um, göl-üm
Lexical exceptions and invariant suffixes remain.

Kazakh

Front/back
Productive alternation

Rounding
Well developed in speech
Standard spelling does not fully record rounding in noninitial syllables.

Uzbek

Front/back
Weaker in many urban varieties

Rounding
No clear-cut standard pattern
Northern and other dialectal patterns differ; harmony is not simply absent everywhere.

Uyghur

Front/back
Suffix alternation

Rounding
Limited and variable
Several suffixes lack rounded variants; spoken patterns and spelling differ.

Qualitative comparison, not a vowel inventory. Source: Johanson (2021), pp. 305-315.